Week 2 Tutorial Notes

▼ Agenda

- Q4 → after ‘Data Modelling’
- Q7 … relatively straightforward ER data modelling exercise
- Q8 → after discussing multivalued attributes
- Q9 … different relationships in ER
- Q11 → first ER for class to try
- Q15 … a substantial ER data modelling exercise (if you have time)

Pre-tute

- Introductions 🙂
- Discourse!!
- Quiz due Friday

Course Outline

<u>Content</u>

- Topic Videos
- Lectures
- Textbooks (OPTIONAL)

<u>Activities</u>

- Tutorials
- Prac exercises
- **Quizzes**
- **Assignments**

- **Exam**

▼ Mark Distribution

Item	Topics	Due	Marks	Contributes to
Quizzes	All topics	Weeks 2,3,4,7,8,10	12%	1,2,3,4,5,6,7,8
Assignment 1	SQL/PLpgSQL	Week 5	13%	3,4
Assignment 2	Python/SQL	Week 10	15%	5
Final Exam	All topics	Exam period	60%	1,2,3,4,5,6,7,8

Contact

- Administrative
  - cs3311@cse.unsw.edu.au
- Technical / Coursework
  - Discourse!!
- If you're unsure, contact me:
  - z5419507@ad.unsw.edu.au

Setup

1. Access the vxdb02 server
   - on cse server:

     ```
     $ ssh vxdb02
     ```

   - else:

     ```
     $ ssh YourUserName@vxdb02.cse.unsw.edu.au
     ```

2. (If first time running)

   ```
   $ 3311 pginit
   ```

3. Tell vxdb02 the source of the psql server and start the server

```
$ source /localstorage/$USER/env
$ p1
```

4. Work with a specific database

```
$ psql SomeDatabase
```

5. Stop PSQL server

```
$ p0
```

After this, look into 'help' command

Data Modelling

Aims to:

- describe what **data** is contained in the database

  (e.g., entities: students, courses, accounts, branches, patients, ...)

- describe **relationships** between data items / objects

  (e.g., John is enrolled in COMP3311, Tom's account is held at Coogee)

- describes **constraints** on data

  (e.g., 7-digit IDs, students can enrol in no more than 3 courses per term)

📖 Try it yourself: tutorial Q4

Entity-Relationship (ER) Data Modelling

ER has three major modelling constructs:

▼ **Attribute**

Data item describing a property of interest

- **Entity**

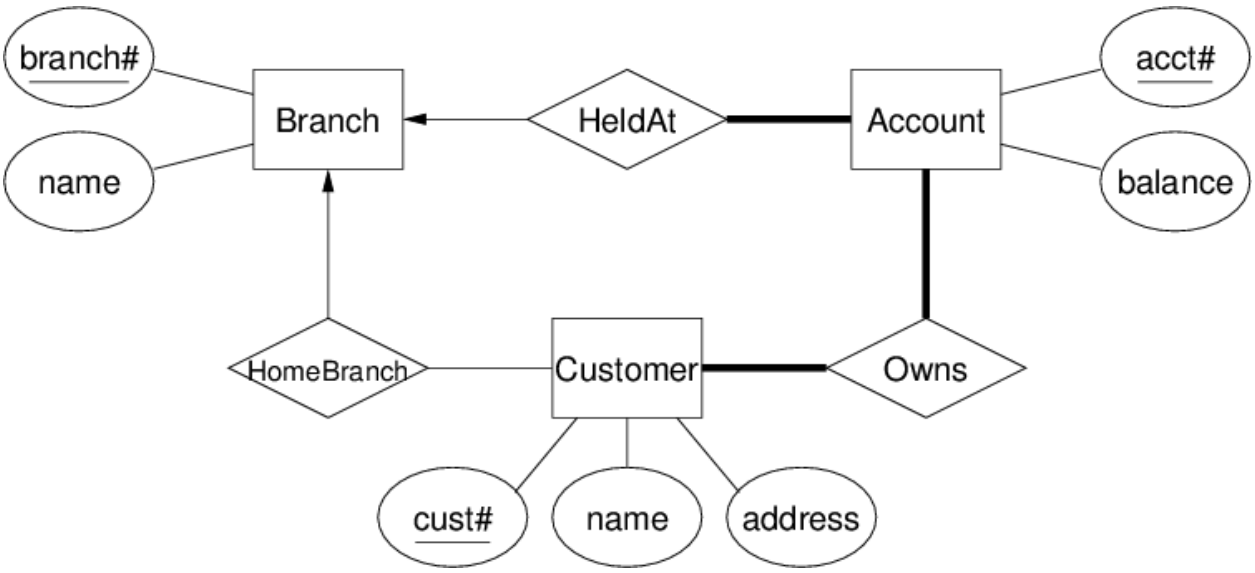
  Collection of attributes describing object of interest

- **Relationship**

  Association between entities (objects)

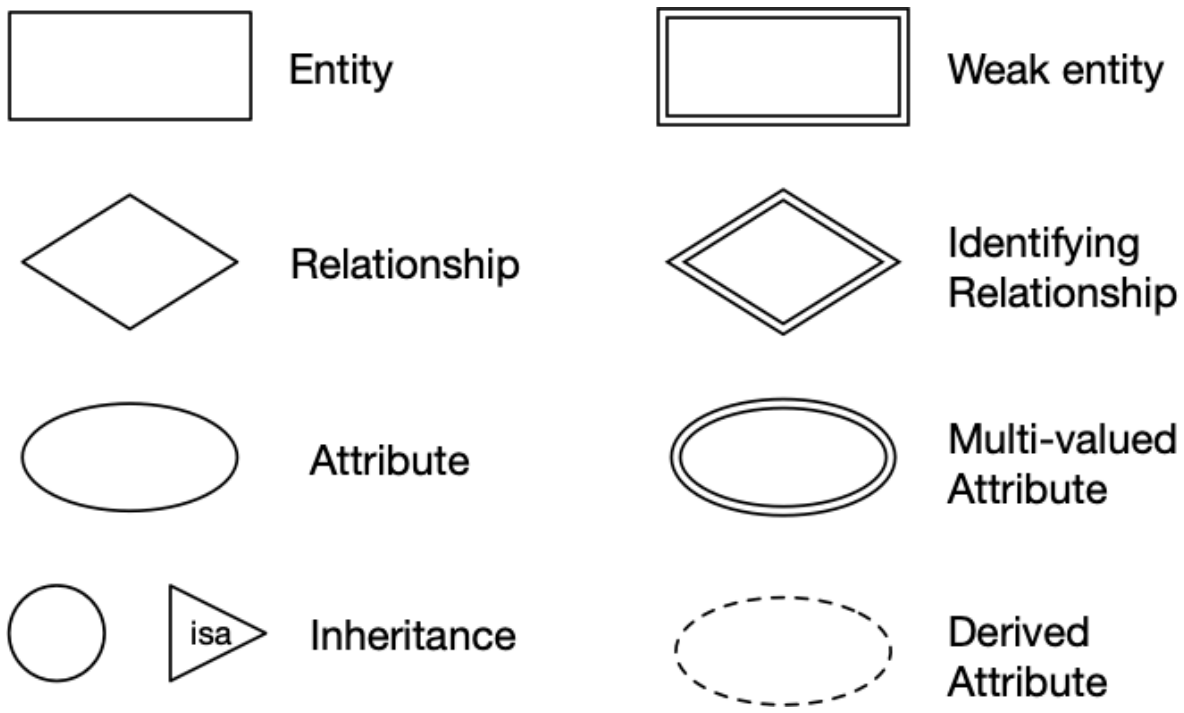
> 👆 Attributes are atomic values. Entities are collections of attributes.

Example: Banking system

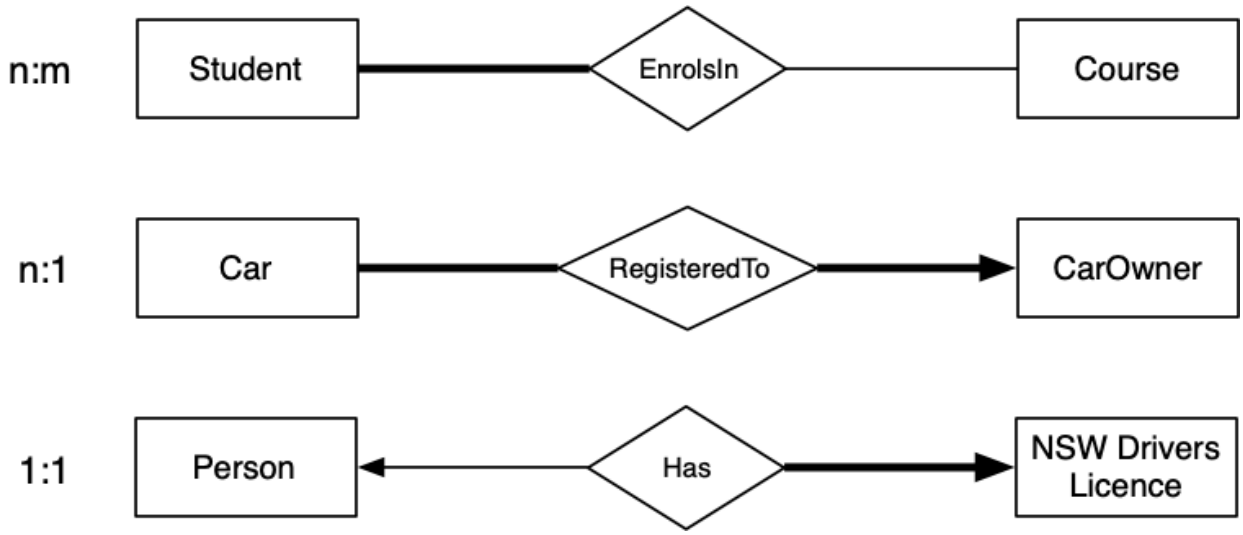


Underline: Primary key

- ER design elements



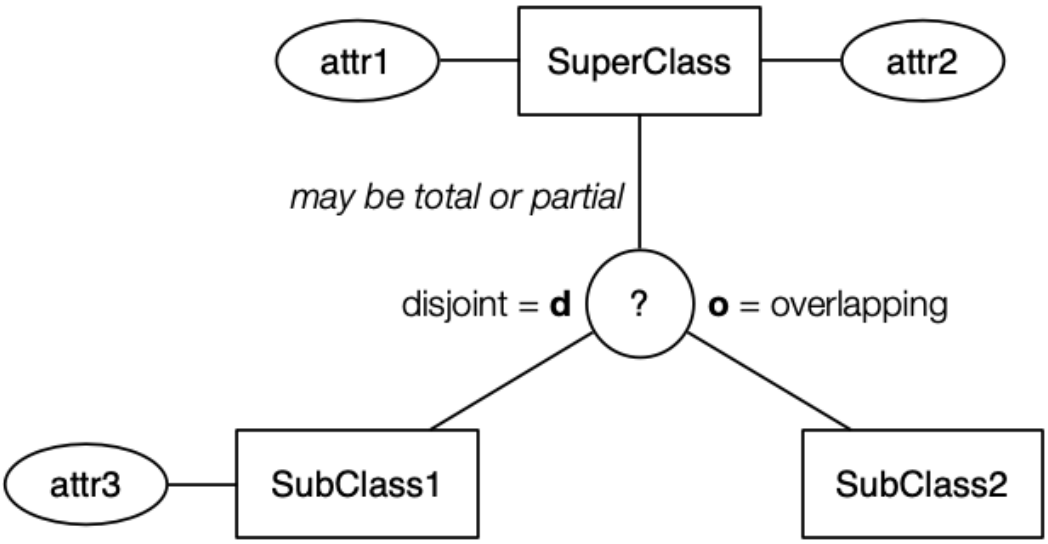
ER Relationships (totality and cardinality)



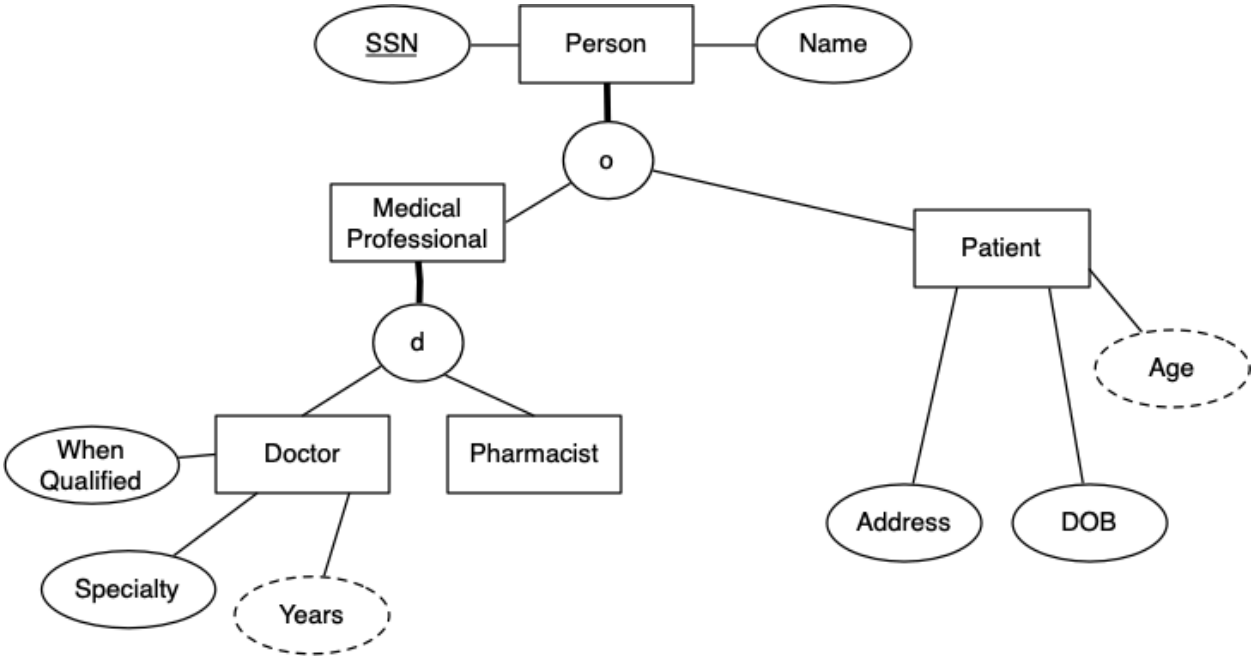
Thick line = total participation; thin line = partial participation; arrow = at most 1

📖 Try: tutorial Q7, Q8, Q9

ER Class Hierarchies



Example: Medical Information



Relational Data Model

A collection of inter-connected **relations** (that look awfully close to tables)

Each **relation** (denoted R,S,T,...) has:

- a name (unique within a given database)

- a set of attributes (which can be viewed as column headings)

Each **attribute** (denoted A,B,... or a_1, a_2,...) has:

- a name (unique within a given relation)
- an associated domain (set of allowed values)

Relation: R($a_1 : D_1, a_2 : D_2, \ \dots \ , a_n : D_n$)

Relational schema: a collection of relation schemas

Tuple of R: an element of $D_1 \times D_2 \times \dots \times D_n$ (i.e. list of values)

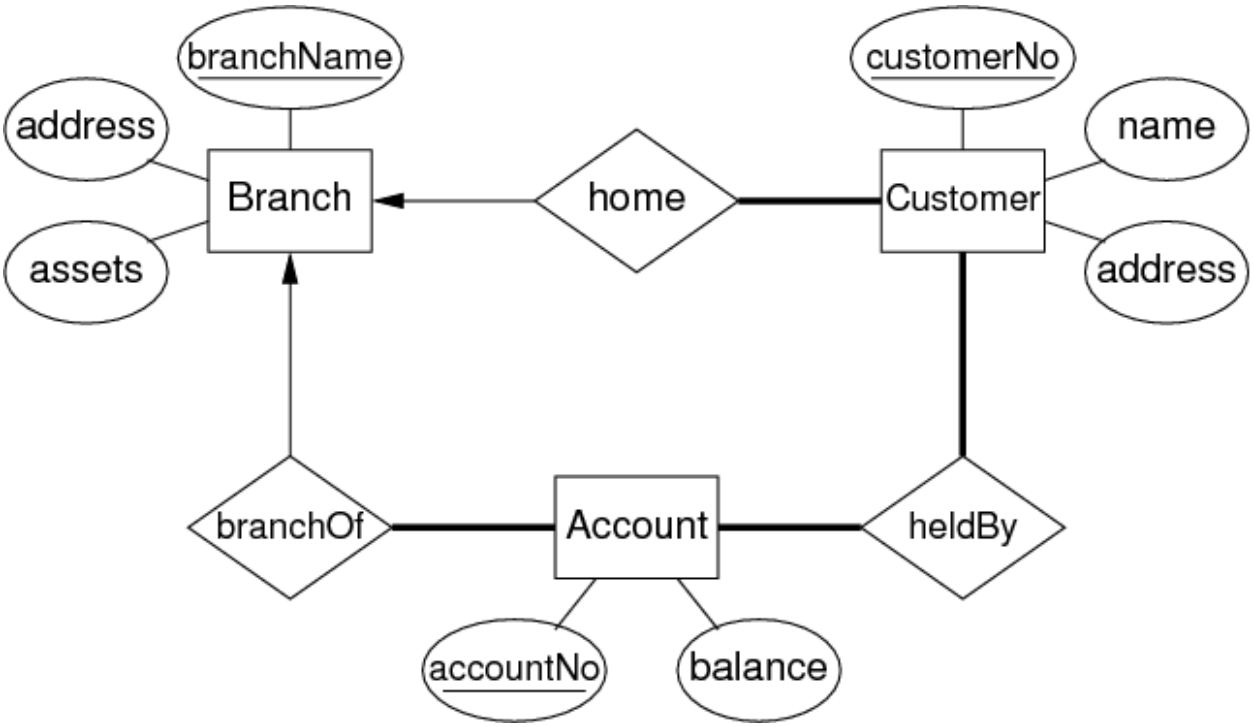
Instance of R: subset of $D_1 \times D_2 \times \dots \times D_n$ (i.e. set of tuples)

Example: Bank Account

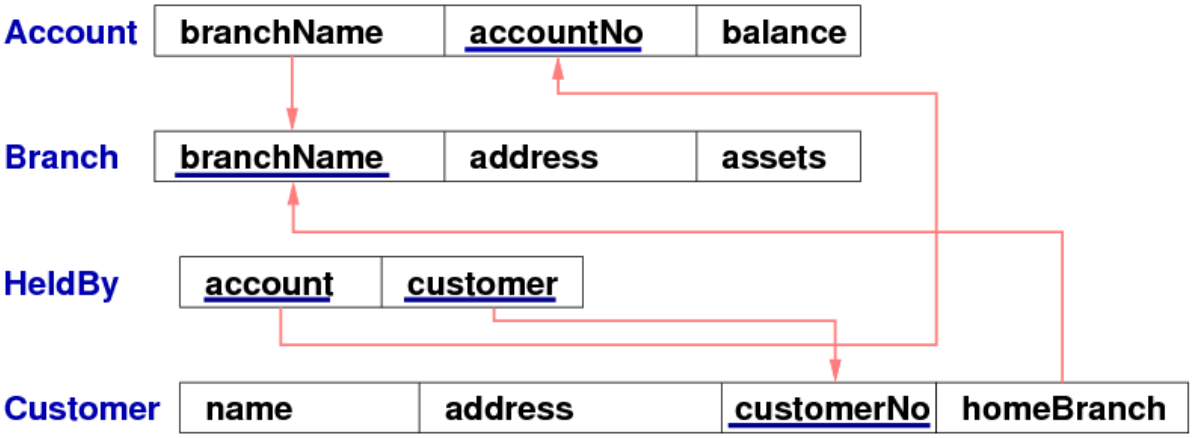
A relation: **Account(branchName, accountNo, balance)**. An *instance* of this relation:

```
{
  (Sydney, A-101, 500),
  (Coogee, A-215, 700),
  (Parramatta, A-102, 400),
  (Rouse Hill, A-305, 350),
  (Brighton, A-201, 900),
  (Kingsford, A-222, 700)
  (Brighton, A-217, 750)
}
```

ER → Relational



▼ Relational. **Identify the Primary and Foreign Keys**



SQL DDL

SQL data definition language (DDL) is the formal way of describing the above relational schemas. The primary SQL DDL construct is table creation

```
create table TableName (
   attr1Name  type [constraints],
   attr2Name  type [constraints],
   attr3Name  type [constraints],
   ...
   primary key (attrxName ),
   foreign key (attryName)
               references OtherTable (attrzName )
);
```

Let’s apply what we’ve learnt!

📖 11, 15